Sri Chaitanya ACADEMY | Infinity Learn



JEE FINAL TEST SERIES

JEE Advanced Part Test - 1_P2 [2023 P1]

Test Date: 10th Apr 2025

INSTRUCTIONS

- Use the OMR sheet provided separately for answering the questions.
- Blank spaces are provided within this booklet for rough work.
- Write your roll number in the space provided on the front cover of this Question Paper
- You are allowed to take away the Question Paper at the end of the examination.
- The OMR will be collected by the invigilator at the end of the examination.
- Do not tamper with or mutilate the OMR. Do not use the OMR for rough work.
- Write your name, roll number and sign with pen in the space provided for this purpose on the OMR.
- Do not write any of these details anywhere else on the OMR. Darken the appropriate bubble under each digit of your roll number.
- Use a BLACK BALL POINT PEN to darken the bubbles on the OMR>
- The OMR is machine-gradable. Ensure that the bubbles are darkened in the correct way.
- Darken the bubbles ONLY IF you are sure of the answer. There is NO WAY to erase or "un-darken" a darkened bubble.

JEE-ADVANCE-2023-P1-Model

Time:3Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 3)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – II(Q.N : 4 – 7)	Questions with Single Correct Choice	+3	-1	4	12
Sec – III(Q.N : 8 – 13)	Questions with Integer Answer Type	+4	0	6	24
Sec – IV(Q.N : 14 – 17)	Matching Type	+3	-1	4	12
Total				**17**	**60**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 18 – 20)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – II(Q.N : 21 – 24)	Questions with Single Correct Choice	+3	-1	4	12
Sec – III(Q.N : 25 – 30)	Questions with Integer Answer Type	+4	0	6	24
Sec – IV(Q.N : 31 – 34)	Matching Type	+3	-1	4	12
Total				**17**	**60**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 35 – 37)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – II(Q.N : 38 – 41)	Questions with Single Correct Choice	+3	-1	4	12
Sec – III(Q.N : 42 – 47)	Questions with Integer Answer Type	+4	0	6	24
Sec – IV(Q.N : 48 – 51)	Matching Type	+3	-1	4	12
Total				**17**	**60**

MATHEMATICS

Max Marks: 60

SECTION – I
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.

For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

---

1. Let A be a 2×2 matrix with rational entries such that $\det\left(A^2-2I_2\right)=0$ where det or $|.|$ represents determinant and I_2 is 2×2 identity matrix. Choose correct option(s).

A) $A^{100}=2^{50}A$ **B)** $A^{99}=2^{49}A$ **C)** $\left|A^{40}\right|=2^{40}$ **D)** $\left|A^{9}\right|=2^{9}$

2. Consider the locus equation

$$(y-z)(x-z)+2(x-y)(x-z)+3(x-y)(y-z)=0\ \ (x,\ y,\ z\in R)$$

representing a pair of plane equation in 3D coordinate system. Let (l,m,n) be the direction cosines along the line of intersection of the pair of planes.

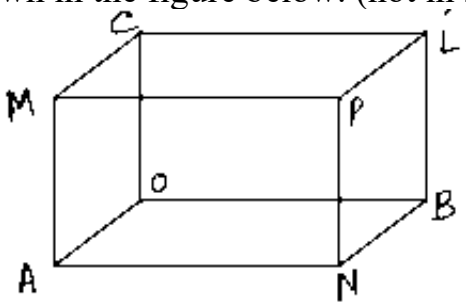
A) Acute angle between the pair of planes is $\frac{\pi}{3}$

B) Angle between the pair of planes is $\frac{\pi}{2}$

C) $2l^2+m^2-3n^2=0$

D) $l^2+2m^2+3n^2=2$

3. OA, OB and OC are coterminous edges of a rectangular parallelopiped and with OA = 2, OB = 4, OC = 3 as shown in the figure below. (not in scale)



Shortest distance between OA and CN is p and between OA and BM is q. The shortest distance p and q divides OA in the ratio $\lambda:1$ and $\mu:1$ respectively.

A) $p=q$ **B)** $p=\frac{12}{5}$ **C)** $\lambda.\mu=1$ **D)** $\lambda=\frac{9}{16}$

SECTION – II
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

---

4. Consider all possible polynomials $P(x) = a_0 + a_1x + a_2x^2 + \ldots\ldots + a_{10}x^{10}$ with real coefficients satisfying $|P(x)| \le |x|$ for all $x \in [-1,1]$. Find the maximum possible value of $(2a_0 + 3a_1)^{10}$.

A) 2^{10} **B)** 3^{10} **C)** 5^{10} **D)** 1

5. Consider 200 integers having mean, median, range and the unique mode all having the same value 200. If A is the largest integer among these 200 integers, find the sum of the digits of the maximum value of A.

A) 21 **B)** 4 **C)** 20 **D)** 2

6. Let $\{a_n\}$ be a sequence satisfying $a_1 = 1$ and $a_n = a_{n-1} + \dfrac{1}{a_{n-1}} \forall\, n > 1, n \in N$. If the value of a_{75} lies in the interval $(3k, 3k+3)$ when $k \in N$, then find the value of k.

A) 5 **B)** 4 **C)** 6 **D)** 3

7. Consider 12×12 square matrix filled with natural numbers from 1 to 144 as given:

$$\begin{bmatrix} 1 & 13 & 25 & \ldots & 133 \\ 2 & 14 & 26 & \ldots & 134 \\ 3 & 15 & 27 & \ldots & 135 \\ 4 & 16 & 28 & \ldots & 136 \\ \vdots & \vdots & \vdots & \ldots & \vdots \\ 12 & 24 & 36 & \ldots & 144 \end{bmatrix}$$

. A number is selected at random from the board and the corresponding row and column are deleted. The process is continued upto 12 times so that no row and no column is left. If sum of all the selected numbers is N, then find the sum of digits of N.

A) 10 **B)** 22 **C)** 15 **D)** 7

SECTION-III

(NON-NEGATIVE INTEGER.)

This section contains **SIX (06)** questions.

- The answer to each question is a **NON-NEGATIVE INTEGER**.
- For each question, enter the correct integer corresponding to the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme

Full Marks : +4 If ONLY the correct integer is entered; Zero Marks : 0 In all other cases.

---

8. Find sum of all possible integer(s) integers m such that the equation $x^3+(m+1)x^2-(2m-1)x-(2m^2+m+4)=0$ has an integer root.

9. If the solution set of the equation $\left[2^x\right]+\left[3^x\right]=\left[6^x\right]$ is given by $x\in(-\infty,0)\cup\left[\log_b^a,\log_b^c\right)\cup\left[\log_c^a,\log_b^d\right)$ where $a,b,c,d\in N$ and [.] denotes the greatest integer function, then the value of $a-b+c+d$ is = ?

10. If $\sin\left(1+\cos^2 x+\sin^4 x\right)=\dfrac{13}{14}$ for some $x\in R,$ find the least integer just more than the value of $10\left(\sin\left(1+\sin^2 x+\cos^4 x\right)-\cos\left(1+\sin^2 x+\cos^4 x\right)\right)$

11. Consider the locus equation $\left[x^2+y^2\right]-x-y=0$ ([.] denotes G.I.F) which consists of line segments when plotted in Cartesian plane. If the sum of lengths of all possible line segment is given by $\sqrt{a}+\sqrt{b}-\sqrt{c}\,(a,b,c\in N)$. find $a+b+c=$

12. Consider the highest power of 5 in the product $1^1.2^2.3^3.....n^n$ is given by $H(n)$ (i.e., $H(n)$ denotes the largest integer k such that 5^k is an integral divisor of above product). Find the value of $\lim\limits_{n\to\infty}\dfrac{n^2}{H(n)}$.

13. Let $a,b,c\in R$ such that $abc+a+c=b.$ If the maximum value of the expression $E(a,b,c)=\dfrac{2}{a^2+1}-\dfrac{2}{b^2+1}+\dfrac{3}{c^2+1}$ is E_m and corresponding values of a,b,c are a_m,b_m and c_m respectively, then find the value of $E_m.b_m\left(a_m+c_m\right).$

SECTION – IV
(MATCHING TYPE)

This section contains **FOUR (04)** Matching List Sets.

- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II.**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks:+3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: –1 In all other cases.

---

14. Consider the family of lines $ax+by+c=0$ where $4a^2+9b^2-c^2+12ab=0$. Let $'p'$ denote distance of $R(6,8)$ from the line $ax+by+c=0$

Match the following columns:

Column – I (Value of p)		Column – II (No. Of lines)	
A)	$p=\sqrt{41}$	P)	1
B)	$p=2/\sqrt{13}$	Q)	2
C)	$p=\sqrt{185}$	R)	3
D)	$p=\sqrt{80}$	S)	4
		T)	5

A) A-P, B-R, C-Q, D-S **B)** A-R, B-P, C-S, D-Q
C) A-R, B-S, C-P, D-Q **D)** A-Q, B-S, C-P, D-R

15.

COLUMN – I		COLUMN – II	
A)	If $\sin\theta=3\sin(\theta+2\alpha)$, then the value of $\tan(\theta+\alpha)+2\tan\alpha$ is	P)	0
B)	If $p\sin\theta+q\cos\theta=a$ and $p\cos\theta-q\sin\theta=b$ then $\frac{p+a}{q+b}+\frac{q-b}{p-a}+1$ is equal to	Q)	1
C)	The value of the expression $\cos\frac{\pi}{7}\cos\frac{2\pi}{7}\cos\frac{10\pi}{7}-\sin\frac{\pi}{14}\sin\frac{3\pi}{14}\sin\frac{5\pi}{14}$ is	R)	$\frac{1}{2}$
D)	For $\alpha=\frac{2\pi}{13}$, $\cos\alpha\cos5\alpha+\cos2\alpha\cos3\alpha+\cos4\alpha\cos6\alpha=$	S)	$-\frac{1}{4}$
		T)	$-\frac{1}{2}$

A) A-R, B-Q, C-P, D-S **B)** A-R, B-S, C-Q, D-P
C) A-P, B-R, C-S, D-Q **D)** A-P, B-P, C-S, D-S

16. Now match the following List – I with List – II.

List – I		**List – II**	
A)	Let $f:R\to R$ be a differentiable function and	P)	0

	$f(1)=1, f'(1)=3;$ then the value of $\lim\limits_{x\to 1}\int\limits_{1}^{x^2}\frac{f(t)-t}{(x-1)^2}dt$		
B)	$\lim\limits_{n\to\infty}\left(\frac{1+\sqrt[n]{4}}{2}\right)^n$ is equal to	Q)	- 1
C)	If $\lim\limits_{n\to\infty}\frac{2x}{\pi}\tan^{-1}(nx); x>0$ then $\lim\limits_{x\to 0^+}\left[f(x)-1\right]$ is {where [.] represents greatest integer function.	R)	2
D)	$\lim\limits_{n\to\infty}\left[\sum\limits_{r=1}^{n}\frac{1}{2^r}\right]$ where [.] represents greatest integer function	S)	4
		T)	1

A) A-S, B-R, C-Q, D-P **B)** A-P, B-T, C-R, D-S
C) A-Q, B-T, C-S, D-R **D)** A-S, B-P, C-T, D-R

17. Match the following Column – I with Column – II.

List – I		**List – II**	
A)	If $2\sin x, \sin 2x$ and $2\cos x$ are in A.P., then value of $\lvert\sin x+\cos x\rvert$ is	P)	$\frac{\sqrt{5}-1}{2}$
B)	If angles of triangle are in A.P. with common difference equal to $\frac{1}{3}$ of the greatest angle then, ratio of the two greater sides is	Q)	$\frac{\sqrt{3}}{2}$
C)	Let the sides AB, BC, CD & DA of a cyclic quadrilateral ABCD are in G.P. with common ratio 2, then $\frac{BD}{AB}$ is	R)	$2\sqrt{5}$
D)	Let a,b,c be in A.P. and a^2,b^2,c^2 be in H.P. If $a\neq c,$ then $a:c$ is	S)	$\sqrt{3}-2$
		T)	1

A) A-Q, B-R, C-P, D-S **B)** A-P, B-Q, C-R, D-S
C) A-P, B-Q, C-S, D-R **D)** A-Q, B-S, C-P, D-R

PHYSICS **Max Marks: 60**

SECTION – I
(ONE OR MORE CORRECT ANSWER TYPE)

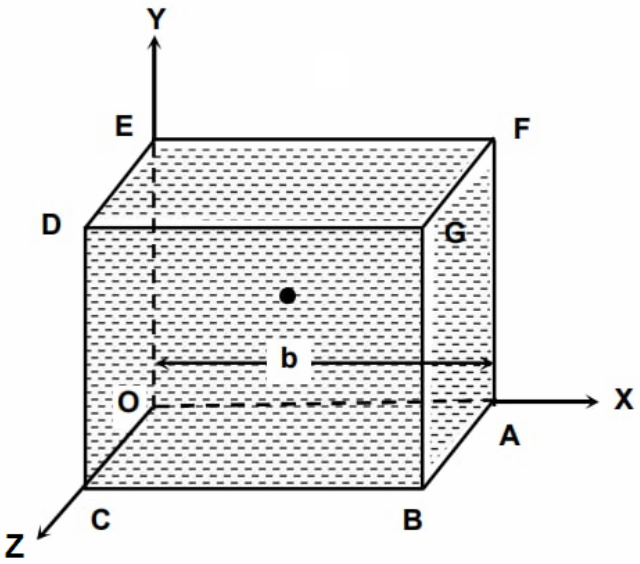
This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.
For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

---

18. A liquid of density $'\rho'$ is filled in a cubical container OABCDEFG of side b. The point 'O' is assumed to be origin. A small ball of density ρ_0 is kept at $\left(\frac{b}{2},\frac{b}{2},\frac{b}{2}\right)$ as shown in figure and an acceleration $2g\hat{i}+g\hat{j}$ is given to container. Uniform gravity exists along negative y-axis. Now, choose the correct option(s).



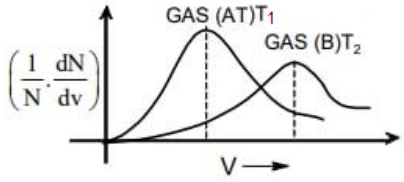
A) If $\rho=\rho_0$ then magnitude of acceleration of small ball is $\sqrt{5}g$.

B) If $\rho=2\rho_0$ then small ball will hit the midpoint of edge GF.

C) If $\rho=3\rho_0$ then small ball will hit the midpoint of edge GF.

D) If $\rho=\rho_0$ then net force on the small ball is zero.

19. According to Maxwell's distribution of molecular speeds, the below graph is drawn for two different samples of gases A and B at temperatures T_1 and T_2 respectively. Which of the following statements is/are INCORRECT ?



A) If $T_1 = T_2$, then molecular mass of gas $B\left(M_B\right)$ is greater than molecular mass of gas $A\left(M_A\right)$.

B) If molecular mass of gas $A\left(M_A\right)$ is equal to molecular mass of gas $B\left(M_B\right)$, then $T_1 > T_2$.

C) If $T_1 < T_2$, then molecular mass of gas $B\left(M_B\right)$ is necessarily less than molecular mass of gas $A\left(M_A\right)$.

D) If gas A is O_2 and gas B is N_2, then considering them to be ideal gases, T_1 is necessarily less than T_2.

20. A car has side window made of glass (shear strength 40 M Pa) having size $40\ cm \times 40\ cm$. The car is travelling with a velocity of $40\ m/s$ in still air. Density of air can be assumed to be $1.25 kg/m^3$. The windows are closed.

A) If the glass is very thin, it may break and fall into the car.

B) If the glass is very thin, it may break & fall out of the car.

C) If we want that glass should not break, it's thickness should be greater than $2.5\ \mu m$.

D) If we want that glass should not break, it's thickness should be greater than $8.75\ \mu m$.

SECTION – II
(SINGLE CORRECT ANSWER TYPE)

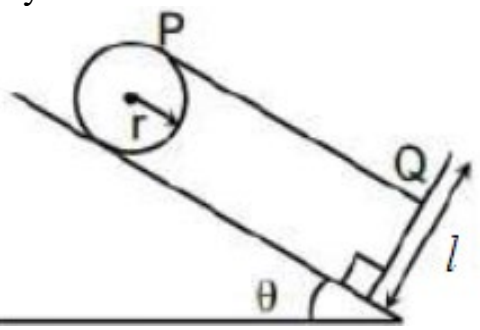
This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

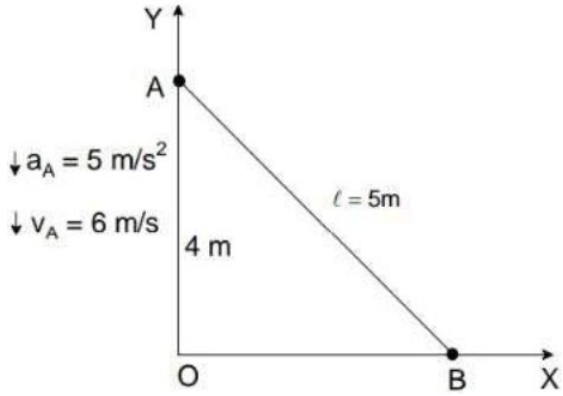
---

21. A uniform rod of length 'l' and a solid cylinder of radius 'r' is kept on a rough inclined plane as shown in the figure. The rod is pivoted, while the cylinder can roll without slipping. A light string PQ attaches top of cylinder to some point on the rod such that string is parallel to the incline. The minimum value of radius of cylinder to ensure that the string is taut when the system is released from rest is



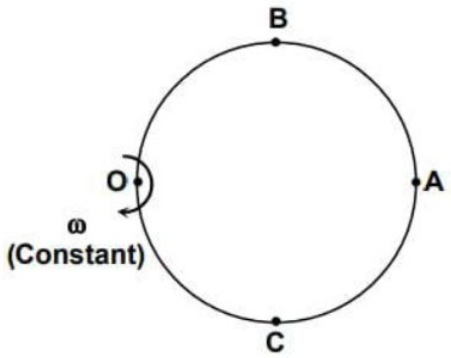
A) $\frac{2}{5}l$ **B)** $\frac{21}{50}l$ **C)** $\frac{4}{9}l$ **D)** $\frac{8}{9}l$

22. Point A on the rod AB has an acceleration of $5\,m/s^2$ and a velocity of $6\,m/s$ at an instant as shown in the figure. The acceleration of the end B at the same moment is



A) $-\frac{80}{3}\hat{i}\,m/s^2$ **B)** $\frac{80}{3}\hat{i}\,m/s^2$ **C)** $-\frac{40}{3}\hat{i}\,m/s^2$ **D)** $40\,\hat{i}\,m/s^2$

23. Suppose a uniform steel ring (mass M and radius R) is kept on a smooth horizontal surface with its plane horizontal. It is hinged about a point O on the perimeter & is made to rotate in a horizontal plane as shown (A, B & C are points on the ring). Then, regarding longitudinal stress developed in the ring :



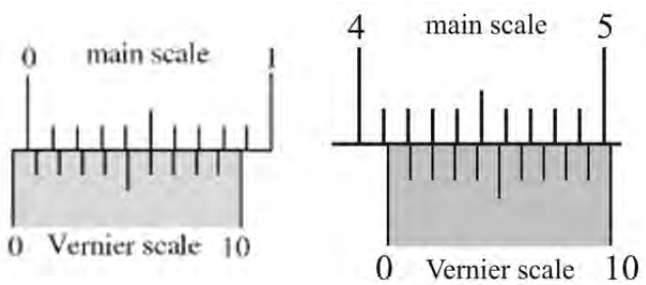
A) it will be maximum at A

B) it will be minimum at A

C) it will be equal at B and C and that value will be maximum.

D) it will be more at O than A by an amount $\frac{M\omega^2}{\pi^2 R}$.

24. The smallest division on the main scale of a Vernier calipers is 0.1 cm. Ten divisions of the Vernier scale correspond to nine divisions of the main scale. The figure below on the left shows the reading of this calipers with no gap between its two jaws. The figure on the right shows the reading with a solid sphere held between the jaws. The correct diameter of the sphere is



A) 4.07 cm **B)** 4.11 cm **C)** 3.15 cm **D)** 4.16 cm

SECTION-III

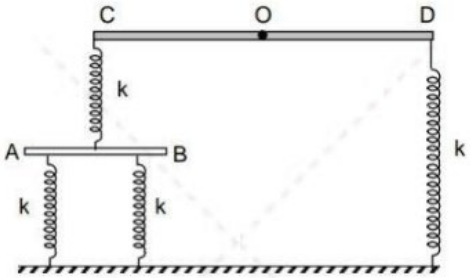
(NON-NEGATIVE INTEGER.)

This section contains **SIX (06)** questions.

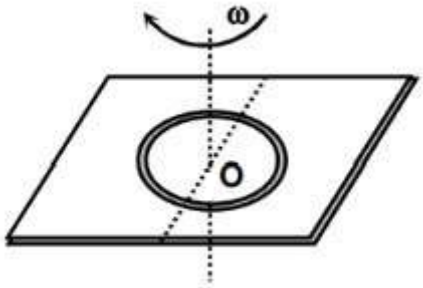
- The answer to each question is a **NON-NEGATIVE INTEGER**.
- For each question, enter the correct integer corresponding to the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme

Full Marks : +4 If ONLY the correct integer is entered; Zero Marks : 0 In all other cases..

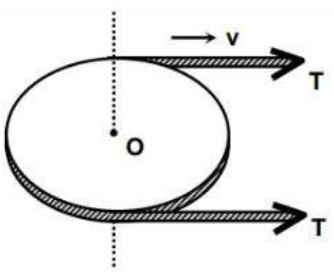
25. In the figure shown, rod AB is light and rigid while rod CD is also rigid and hinged at its midpoint O. Rod can rotate without friction about O. The time period of small oscillation of rod CD is $T = 2\pi\sqrt{\frac{m}{nk}}$ where m is mass of rod CD and k is spring constant of each spring. The value of n is



26. A uniform ring of mass m is placed on a rough horizontal fixed surface as shown in the figure. The coefficient of friction between the left part of the ring and horizontal surface is $\mu_1 = 0.6\pi$ and between right half and the surface is $\mu_2 = 0.2\pi$. At the instant shown, the ring has been imparted an angular velocity about a vertical axis passing through center in clockwise sense in the figure shown. At this moment, magnitude of acceleration of centre O of the ring $\left(in\, m/s^2\right)$ is $\left(\text{take } g = 10\, m/s^2\right)$

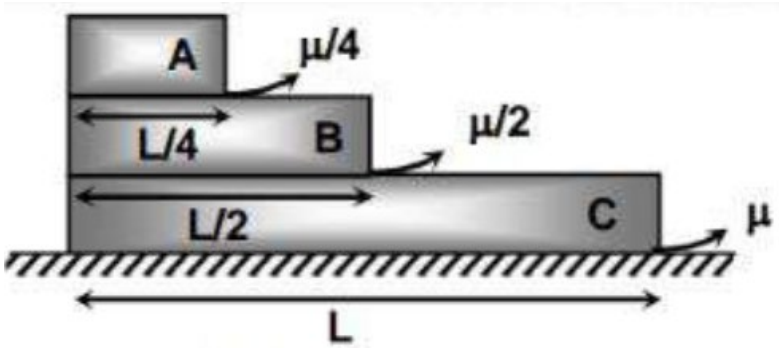


27. A flexible drive belt runs over a frictionless pulley as shown in figure. The pulley is rotating freely about the vertical axis passing through the centre O of the pulley. The vertical axis is fixed on the horizontal smooth surface. The mass per unit length of the drive belt is 1 kg/m and the tension (T) in the drive belt 8 N. The speed(constant) of the drive belt is 2 m/s. Find the net normal force applied by the belt on the pulley in newtons.



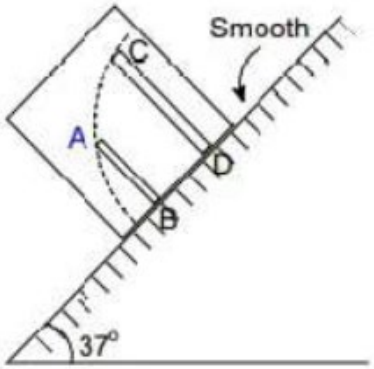
28. Three blocks A, B and C of masses $m, \frac{m}{2}$ and m respectively, of different densities and dimensions are placed over each other as shown in the figure. The coefficients of friction are shown. Blocks are made to move towards right with same velocity at the instant shown and left. Find the time (in s) taken by the upper block A to topple from the middle block B. Assume that blocks B and C don't stop sliding before A topples from B.

$\left(given\, L = 36m, \mu = 0.4\, and\, g = 10\, m/s^2\right)$

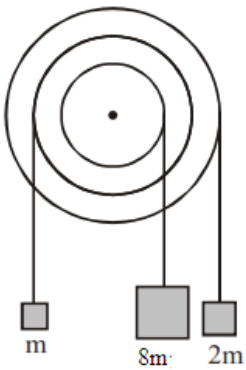


29. Two plates AB and CD are rigidly attached perpendicular to base of a massive box. Box is released on smooth inclined plane having angle of inclination 37°. With what minimum speed (in m/s) with respect to box, a particle is to be projected from base of box so as to graze top points A and C of plates ? $AB = 2m, CD = 3m.$ Given separation between tops of plates is 3m.

Take $g = 10\, m/s^2.$



30. Three weightless pulleys with radii R, 2R and 3R are concentrically fastened together to make a single triple pulley system and mounted on an axis on which the triple pulley can rotate without friction (figure). The threads are light and they do not slip on the pulley. We suspended masses 8m, m and 2m as shown. What is the acceleration $\left(in\, m/s^2\right)$ of mass 8m ? $\left(g = 10\, m/s^2\right)$



SECTION – IV
(MATCHING TYPE)

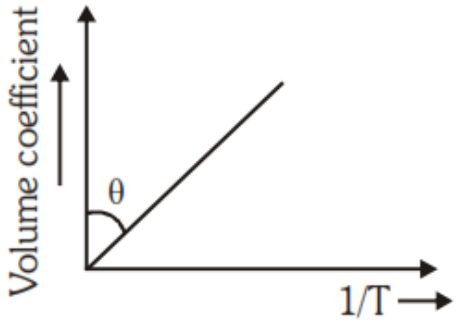
This section contains **FOUR (04)** Matching List Sets.

- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II.**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks:+3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: –1 In all other cases.

---

31. A monoatomic ideal gas of 2 moles was found to be undergoing a thermodynamic polytropic process whose volume coefficient is varying with temperature as shown. $\left(R = \frac{25}{3} J/mol.K\right)$. Match the two columns.



LIST-I		LIST-II	
(P)	Work done by gas in joules, for every 1K rise in temperature, if $\tan\theta = 3$	(1)	0
(Q)	Internal energy rise of the gas in joules, for every 1K rise in temperature, if $\tan\theta = 2$	(2)	25
(R)	Heat absorbed by the gas in joules, for every 1K rise in temperature, if $\tan\theta = 1$	(3)	$\frac{125}{3}$
(S)	Value of θ in degrees for which the internal energy rise of the gas for every 1 K rise in temperature is 25 J	(4)	$\frac{50}{9}$

A) $P \to 4; Q \to 2; R \to 3; S \to 1,2,3,4$

B) $P \to 2,3; Q \to 1,3,4; R \to 2; S \to 3,4$

C) $P \to 4; Q \to 3,4; R \to 2,3,4; S \to 1,2$

D) $P \to 2,4; Q \to 1,2; R \to 4; S \to 1,3,4$

32. Three ants are sitting at the vertices of an equilateral triangle of side a. At t = 0, ant 1 starts approaching ant 2 with a speed v, ant 2 starts approaching ant 3 with a speed v and ant 3 starts approaching ant 1 with a speed v. If a = 1m, v = 1m/s, match the following.

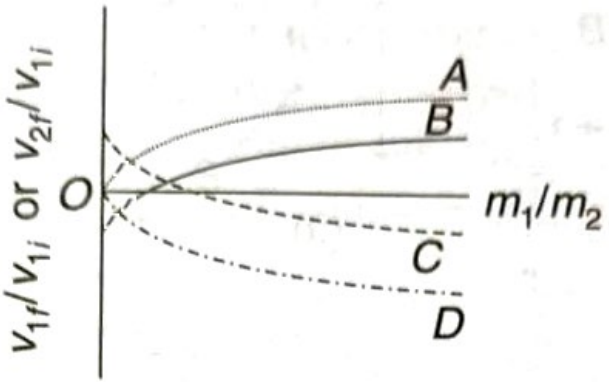
LIST-I	LIST-II

(P)	The three ants meet at the centroid of the triangle at time (in s)	(1)	$\frac{\sqrt{3}}{2}$
(Q)	The angular velocity (in rad/s) of ant 2 with respect to ant 1 at t = 0 is	(2)	$\frac{25\sqrt{3}}{256}$
(R)	The angular acceleration (in rad/s^2) of ant 2 with respect to ant 1 at t = 0 is	(3)	$\frac{2}{3}$
(S)	The area of the equilateral triangle (in m^2) formed by the three ants at time t = a/4v will be	(4)	$\frac{3\sqrt{3}}{4}$

A) $P \to 1; Q \to 4; R \to 2; S \to 3$ **B)** $P \to 3; Q \to 1; R \to 4; S \to 2$

C) $P \to 4; Q \to 3; R \to 1; S \to 2$ **D)** $P \to 3; Q \to 1; R \to 1; S \to 4$

33. A particle of mass m_2 is at rest. Another particle of mass m_1 hits it. Collision is one-dimensional and elastic. Suffix 'f' represents final and 'i' represents initial in the following.



List-I		List-II	
(a)	The curve in the figure, which is the graph of the velocity ratio v_{1f} / v_{1i} versus the mass ratio m_1 / m_2	(p)	A
(b)	The value of the intercept of the graph v_{1f} / v_{1i} versus the mass ratio m_1 / m_2 with the vertical axis is	(q)	C or D
(c)	The maximum value taken by the graph of v_{2f} / v_{1i} versus m_1 / m_2 is	(r)	B
(d)	The curve that is the graph of the velocity ratio v_{2f} / v_{1i} versus m_1 / m_2 is	(s)	0
		(t)	-1
		(u)	2

A) a – r; b – t; c – u; d - p **B)** a – p; b – q; c – r; d – s

C) a – q; b – r; c – s; d - t **D)** a – p; b – t; c – q; d - r

34. A particle of mass $m = 1kg$ can move under influence of a conservative force. The potential energy U of the particle under this force varies with its co-ordinate x as $U = -4(x-1)^2(x-2)^2$ where all quantities are in SI units. Column I gives few initial states of the particle, and Column II lists the possibilities regarding its motion. Match them.

List-I		List-II	
(A)	Particle released at $x = -1m$	(P)	Will oscillate with an angular frequency $2rad/s$.
(B)	Particle released at $x = +1m$	(Q)	Will move in negative x direction
(C)	Particle kept at $x = +1.5m$ and given a very small push in positive x direction	(R)	Will remain in equilibrium
(D)	Particle kept at $x = +1.5m$ and given a very small push in negative x direction	(S)	Will move to infinity

A) A – P, B – Q, C – S, D - R **B)** A – QS, B – R, C – P, D – P

C) A – PQ, B – RS, C – P, D - R **D)** A – Q, B – P, C – S, D – R

CHEMISTRY

Max Marks: 60

SECTION – I
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.
For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

35. Which of the following statements are correct

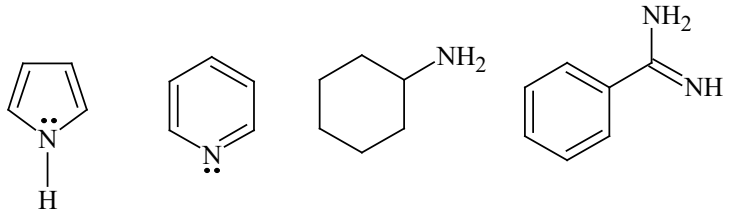
A) Ionic product of heavy water (D_2O) is more than ordinary water (H_2O)

B) If the pD of D_2O is 7.2, it is acadic in nature

C) Since O – D bond is stronger than O – H bond ionization of D_2O is less than H_2O

D) In neutral condition the pH and pD of H_2O and D_2O are equal.

36.



I II III IV

Which of the following statement/s is are correct?

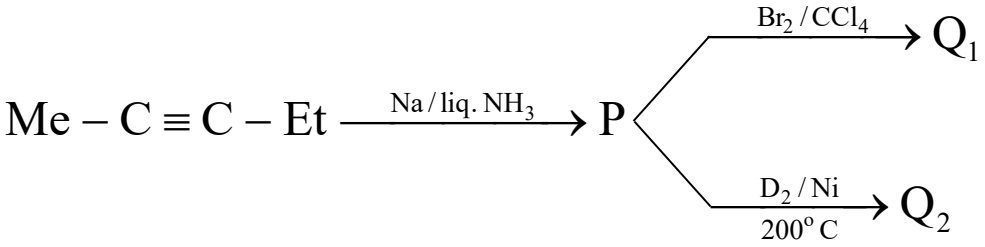
A) I & II are aromatic and have equal basic strength

B) I is aromatic, II is antiaromatic but II is stronger base than I

C) Basicity order of the above compounds is I < II < III < IV

D) Conjugate acid of IV is more stabilized than conjugate acid of II.

37. Some statements regarding the reaction:



are given below. Select the correct statement(s).

A) P is a trans-alkene

B) Q_1 is a pure compound and optically inactive due to internal compensation

C) In the P to Q_1 conversion step the Br_2 adds on P in a syn manner and the intermediate formed is a cyclic brominium ion.

D) Q_2 is a binary mixture and is optically inactive due to external compensation.

SECTION – II
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

38. The table shows the melting points and boiling points of four substances at 1 atm pressure

Element	Melting point (ºC)	Boiling point (ºC)
W	-189	-186
X	-110	-140
Y	-7	60
Z	-9	10

Which substance cannot exist as a liquid at -100ºC and 1 atm pressure

A) W **B)** X **C)** Y **D)** Z

39. The correct statement about C_{60} is

A) C_{60} is soluble benzene

B) C_{60} do not react with alkali metals

C) C_{60} is made up of 10 five membered and 15 six membered rings

D) Two adjacent five-membered rings share a common edge

40. For a gas which of the following statements is incorrect

A) Inversion temperature is double the Boyle temperature

B) Boyle temperature is double the inversion temperature

C) Critical temperature is less than inversion and Boyle temperature

D) Inversion temperature is more than Boyle temperature and critical temperature

41. Enthalpy is equal to :

A) $T^2\left[\frac{\partial(G/T)}{(\partial T)}\right]_P$ **B)** $-T^2\left[\frac{\partial(G/T)}{(\partial T)}\right]_P$ **C)** $T^2\left[\frac{\partial(G/T)}{(\partial T)}\right]_V$ **D)** $-T^2\left[\frac{\partial(G/T)}{(\partial T)}\right]_V$

SECTION-III

(NON-NEGATIVE INTEGER.)

This section contains **SIX (06)** questions.

- The answer to each question is a **NON-NEGATIVE INTEGER**.
- For each question, enter the correct integer corresponding to the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme

Full Marks : +4 If ONLY the correct integer is entered; Zero Marks : 0 In all other cases.

---

42. How many of the following compounds contain 3-centered-2electro bonding

$(BeCl_2)_n, [Be(CH)_3]_n, [BeH_2]_n, (AlH_3)_n, Al_2(CH_3)_6, Al_2Cl_6, B_2H_6, Be(BH_4)_2, Al(BH_4)_3$

43. How many of the following anions do not exist as hydrated ions in water

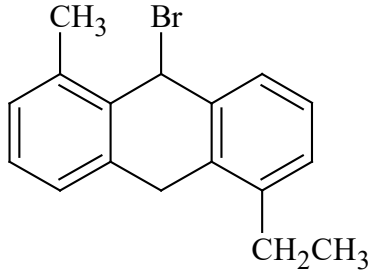
$O^{2-}, O_2^{2-}, O_2^-, O_3^-, N^{3-}, C_2^{2-}, C_3^{4-}, F^-, Cl^-, OH^-, SO_4^-, NO_3^-$

44. Equilibria, $X \rightleftharpoons 2Y$ and $Z \rightleftharpoons P + Q$ are established with 1 mole of X and Z taken initially in separate vessels with equilibrium pressures of 1 atm and 36 atm respectively If the degree of dissociations are same, then $K_{p_2} = nK_{p_1}$. What is the value of 'n' ?

45. Find the quantum number 'n' corresponding to the excited state of He^+ ion, if one transition to the ground state that ion emits two photons in succession with wavelengths 108.5 nm and 30.4 nm.

46. A hydrocarbon (X) contains 91.2% carbon and 8.8% hydrogen. The compound on chlorination using Cl_2/hv and $Cl_2/AlCl_3$ gives three isomeric monochloro substituted products. Find the value of 'Y' where $Y = \frac{\text{Total number of atoms present in (X)}}{5}$

47. What is the total number of resonating structures that are possible for the given compound ?



SECTION – IV
(MATCHING TYPE)

This section contains **FOUR (04)** Matching List Sets.

- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II.**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks:+3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: –1 In all other cases..

---

48. Match the molecular species in List-I with their shape in List-II

List-I		List-II	
A)	Linear shape	P)	CS_2
B)	sp hybridization	Q)	XeF_2
C)	sp^3d hybridization	R)	C_2H_2
D)	CO_2 is isostructural to	S)	NCO^-

A) A-PQRS, B-PRS, C-Q, D-PQRS **B)** A-PQR, B-PS, C-QS, D-PQR
C) A-PQRS, B-PR, C-Q, D-PQ **D)** A-PR, B-PS, C-P, D-S

49. Given two mixtures:

I) $NaOH + Na_2CO_3$

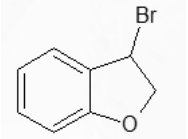
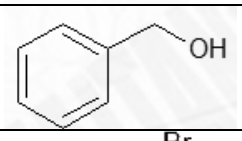
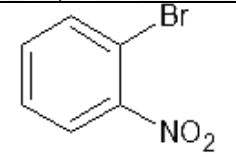
II) $NaHCO_3 + Na_2CO_3$

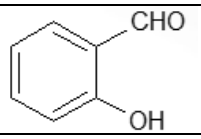
100 ml of mixture I required 'W' and 'X' ml of 1 M HCl in separate titrations using phenolphthalein and Methyl orange indicators. While 100 ml of mixture II required 'Y' and 'Z' ml of same HCl solution in separate titration using same indicators.

Column I (Substance)		Column II (Molarity in solution)	
(A)	Na_2CO_3 in mixture I	**(P)**	$(2w-x)\times10^{-2}$
(B)	Na_2CO_3 in mixture II	**(Q)**	$(z-2y)\times10^{-2}$
(C)	$NaOH$ in mixture I	**(R)**	$y\times10^{-2}$
(D)	$NaHCO_3$ in mixture II	**(S)**	$(x-w)\times10^{-2}$

A) A-S; B-R; C-Q; D-P **B)** A-S; B-R; C-P; D-Q
C) A-P; B-Q; C-S; D-R **D)** A-P; B-S; C-Q; D-R

50. Match each of the compounds given in column-I with the reaction that they can undergo given in column –II

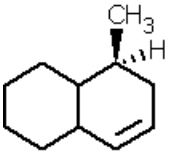
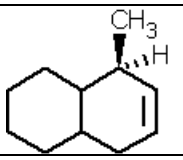
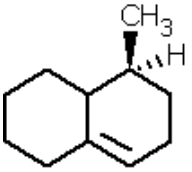
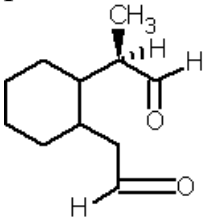
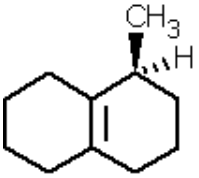
List-I		List-II	
A)		P)	Nucleophilic substitution
B)		Q)	Elimination
C)		R)	Nucleophilic addition

D)		S)	Esteri fi cation with acetic anhydride
		T)	Dehydrogenation

A) A-PQ; B-ST; C-Q; D-RS **B)** A-QT; B-PSR; C-R; D-QR

C) A-PQT; B-PQST; C-P; D-RS **D)** A-PQRST; B-PS; C-QRS; D-RS

51. Match the Following:

List-I		**List-II**	
A)	When reacts with Br_2 / CCl_4 Diastereomer will formed	P)	
B)	When react with O_3 / H_2O_2 Dicarboxylic acid will formed	Q)	
C)	Optically active compound	R)	
D)	When undergo reductive ozonolysis following product will formed 	S)	

A) A-PQ; B-ST; C-Q; D-R **B)** A-PQT; B-PQST; C-P; D-RS

C) A-QT; B-PSR; C-R; D-Q **D)** A – PQRS; B – PQ; C– PQRS; D – Q